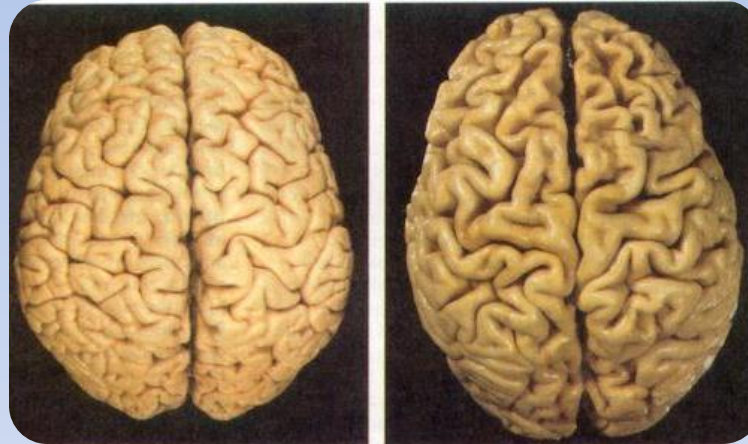


ALZHEIMER'S DISEASE



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Outlines

- **What is Alzheimer disease?**
- **Introduction**
- **Clinical features**
- **Relative risk factors**
- **Histopathology & pathophysiology**
- **Genetics**
- **Diagnosis**
- **Neuroimaging**
- **Treatment of dementia**
- **Take home message**

Alzheimer's Disease (AD)

- AD is a neurodegenerative disorder marked by cognitive and behavioral impairment that significantly interferes with social and occupational functioning.
- It is an incurable disease with a long preclinical period and slowly progressive course.

Introduction Epidemiology

- **AD is the most common dementia (65% of all dementias).**
- The incidence increases exponentially with age over 65 years (5%).
- **Late-onset AD** affects **up to 20% of those >80 y.**
- **Early-onset AD** (onset < 65 y) is unusual, and in some but not all cases is familial.
- Familial early-onset AD makes up < 1% of cases and often follows an **autosomal-dominant** pattern of inheritance.
- The average **life expectancy** after diagnosis has been reported to be **8 -10 y** but may range from 3 - 20 y.

Deficits occur in the following domains:

- **Memory:** forgetfulness
- **Disorientation:** confusion about the location of familiar places
- **Speed of performance:** taking longer to accomplish normal, daily tasks
- **Problem solving:** trouble handling money and paying bills
- **Language:** problems with reading, writing, working with numbers end with aphasia.
- **Perceptual-motor problems:** such as trouble getting out of a chair or setting the table
- **Judgment:** compromised judgment, often leading to bad decisions

Clinical Features of AD

- Cognitive deficits in AD appear & progress insidiously.
- **Memory** impairment, mainly recent events, is usually its **first manifestation**.
- Deficits in other cognitive domains may appear with or after the development of memory impairment.
- **Executive** dysfunction and impaired **visuospatial** skills tend to be affected relatively in **early stage**, while deficits in **language** function and **behavioral** symptoms often manifest **later in the disease course**.
- Neuropsychiatric & behavioral symptoms are common in **middle & late stages** of AD, but may occur early in the course of the illness in some patients.
- **Non-cognitive** neurologic deficits (**pyramidal & extrapyramidal** motor signs, **myoclonus**, & **seizures**) can occur in **late stages** of AD but are uncommon in early & middle stages.

Atypical AD

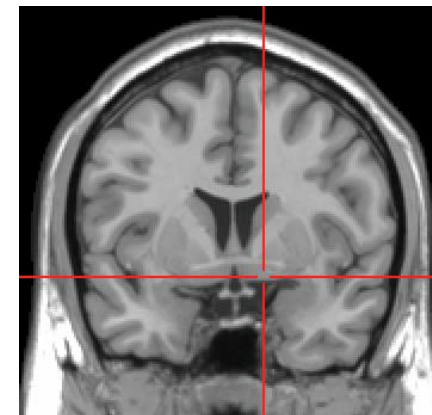
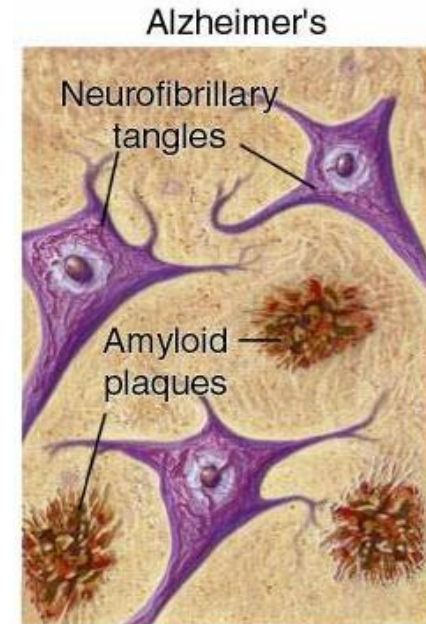
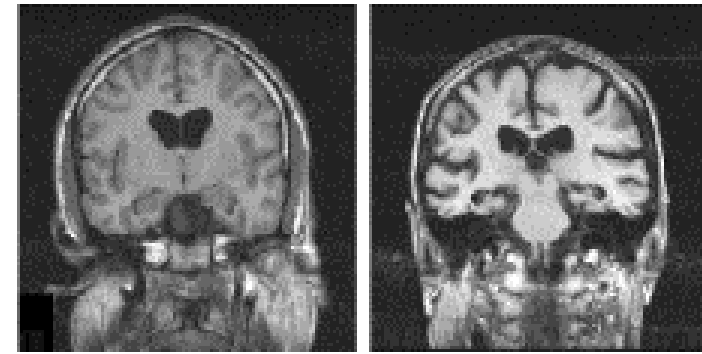
- **Atypical** presentations of AD have been described. These include:
 - » Visual variant (posterior cortical atrophy),
 - » Progressive aphasia, &
 - » Progressive executive dysfunction as the predominant symptom.
- Atypical presentations are more common in **younger** people with AD.
- **Late-onset AD** should be suspected in any older adult with insidious onset, progressive decline in memory, and at least one other cognitive domain leading to impaired functioning.

RELATIVE RISK FACTORS

- **Family history of:**
 - Dementia
 - Down syndrome
 - Parkinson's disease
- **Maternal age > 40 years**
- **History of:**
 - Head trauma (with LOC)
 - History of depression
 - History of hypothyroidism
 - History of severe headache

Histopathology of AD

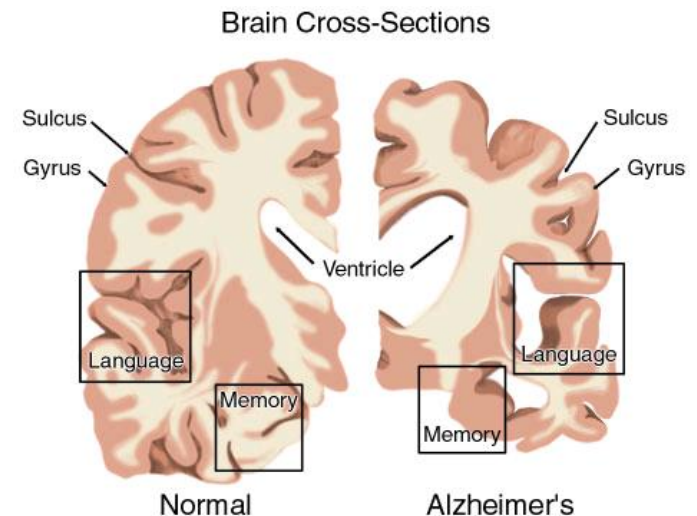
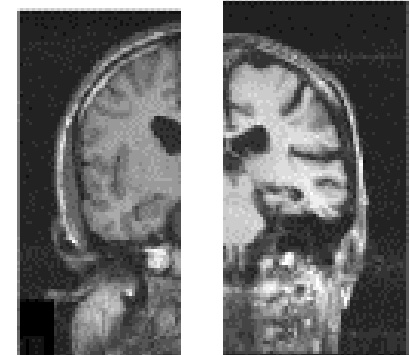
- Generalized cortical neuronal loss (Brain Atrophy) >>>
- Senile plaques & neurofibrillary tangles (with amyloid composed of β -A4 protein), are precipitated leading to granulo-vacuolar degeneration >>>
- Neuronal loss in nucleus basalis of Meynert >>>



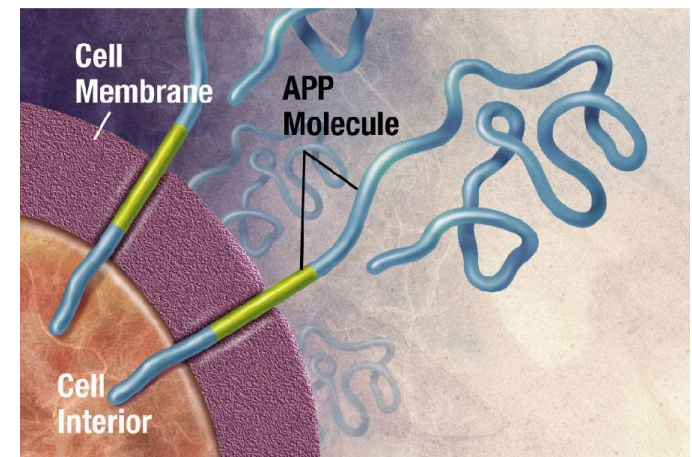
Pathophysiology

Abnormalities & loss of neurons in hippocampus, cerebral cortex & basal forebrain.

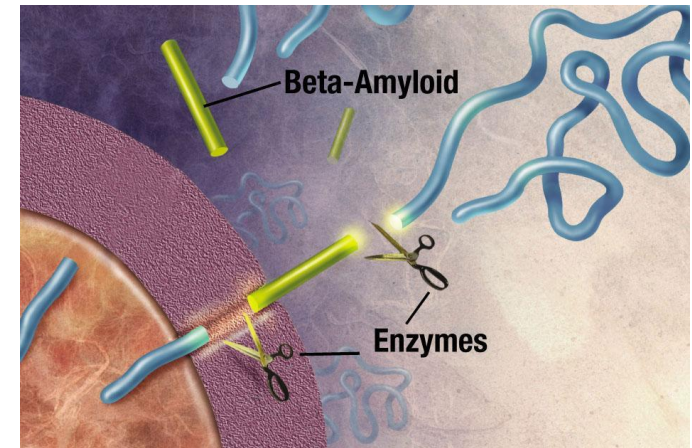
- **Marked decrease in cortical** choline acetyltransferase (**ChAT**) and acetylcholine (**ACh**) in the cerebral cortex and nucleus basalis of Meynert (**nbM**)
- **Degree of cholinergic deficit correlates with degree of dementia.**



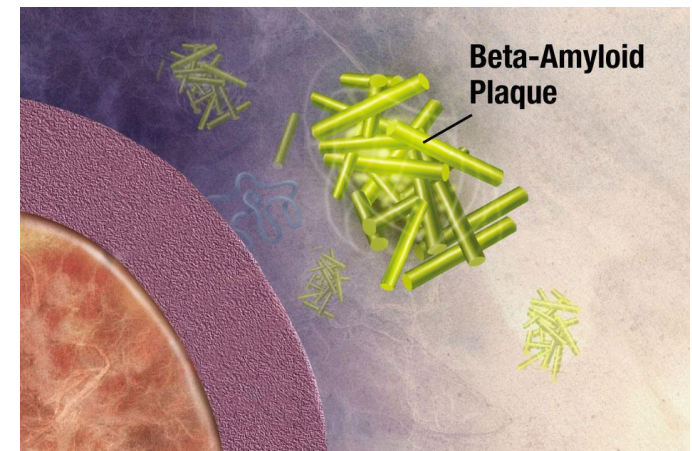
A) **APP** sticks through the neuron's membrane, partly inside and partly outside the cell.



B) **Enzymes** act on the APP and cut it into fragments of protein, one of which is called **beta-amyloid**.

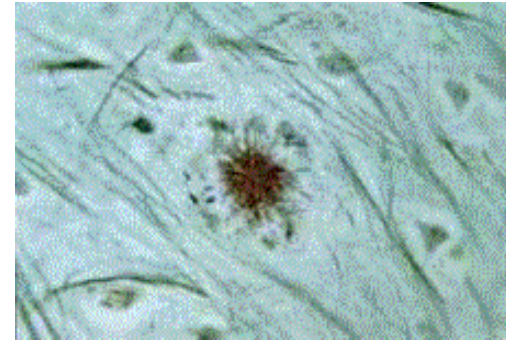


C) The beta-amyloid fragments begin coming together into clumps outside the cell, then join other molecules and non-nerve cells to form **insoluble plaques**.



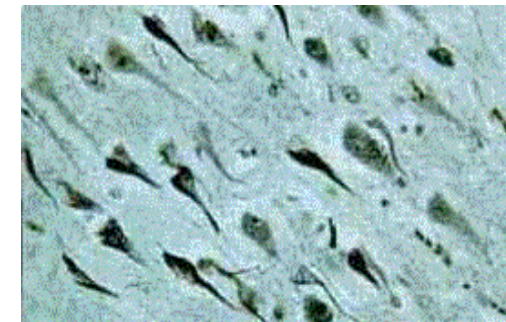
Senile Plaques

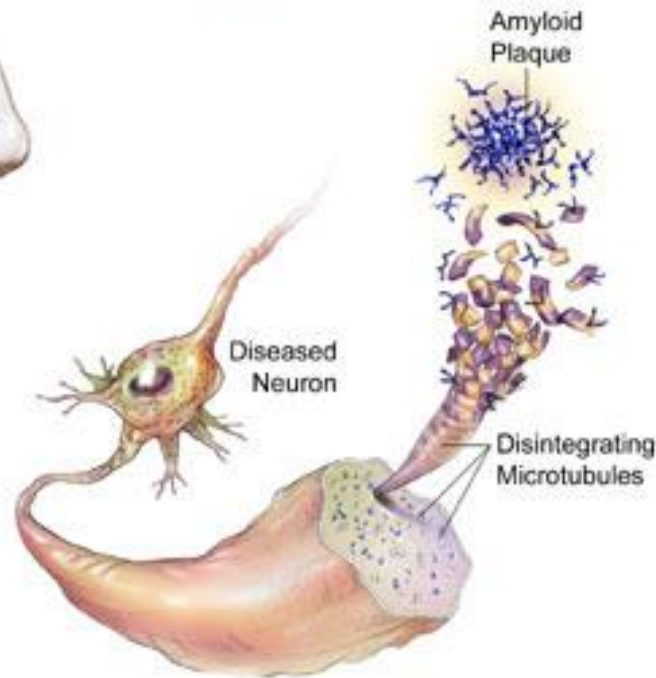
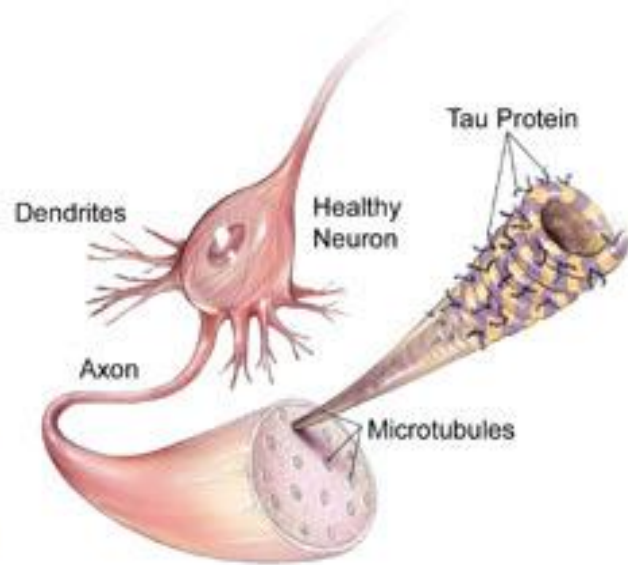
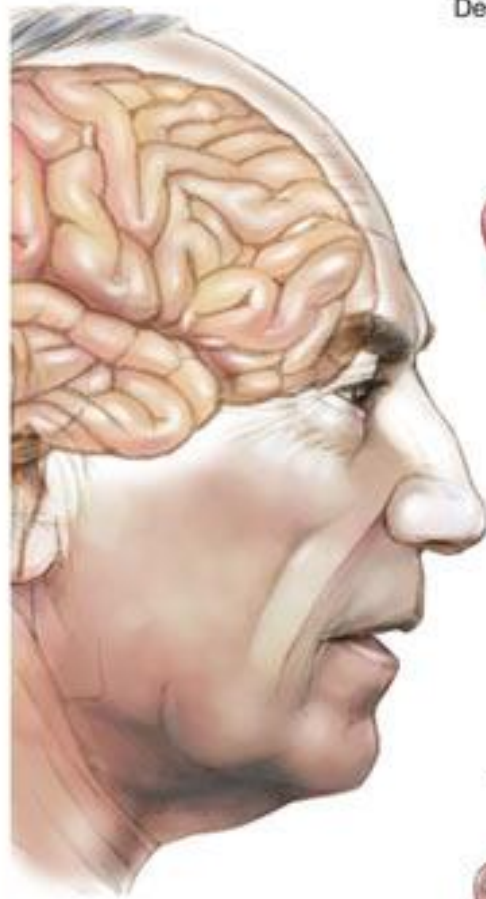
- **Extracellular** deposits of amyloid- β peptide forming plaque core surrounded by axons/dendrites & glia
 - Small number of plaques found in **normal** aging brain
- Amyloid- β peptide derived from amyloid precursor protein (APP)
 - APP gene is located on chromosome **21**
 - Mutations in APP gene give rise to some familial forms of AD



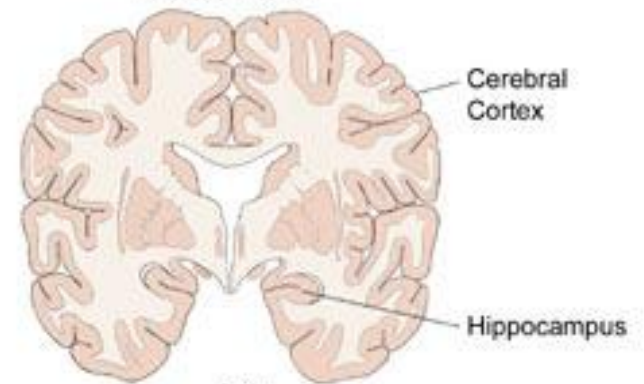
Neurofibrillary Tangles

- Intra-neuronal fibers composed of paired helical filaments of tau protein that is hyper-phosphorylated
 - Tau protein is **normally** found in **axons** associated with microtubules
 - In AD: tau protein is self-aggregates & found **throughout the neuron**

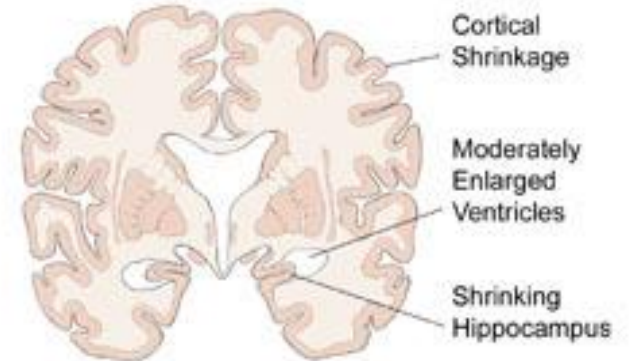




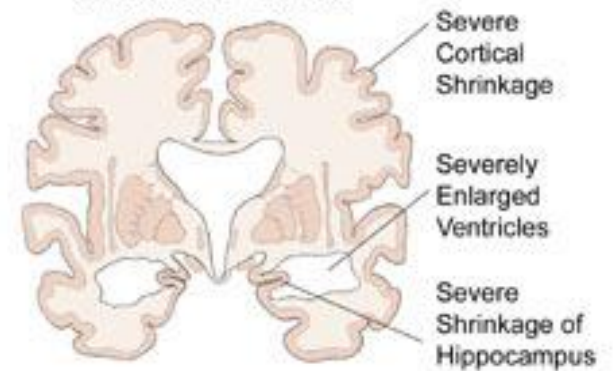
Healthy Brain



Mild Alzheimer's Disease



Severe Alzheimer's Disease



Healthy Neuron

Microtubules

Stabilizing
Tau Molecules

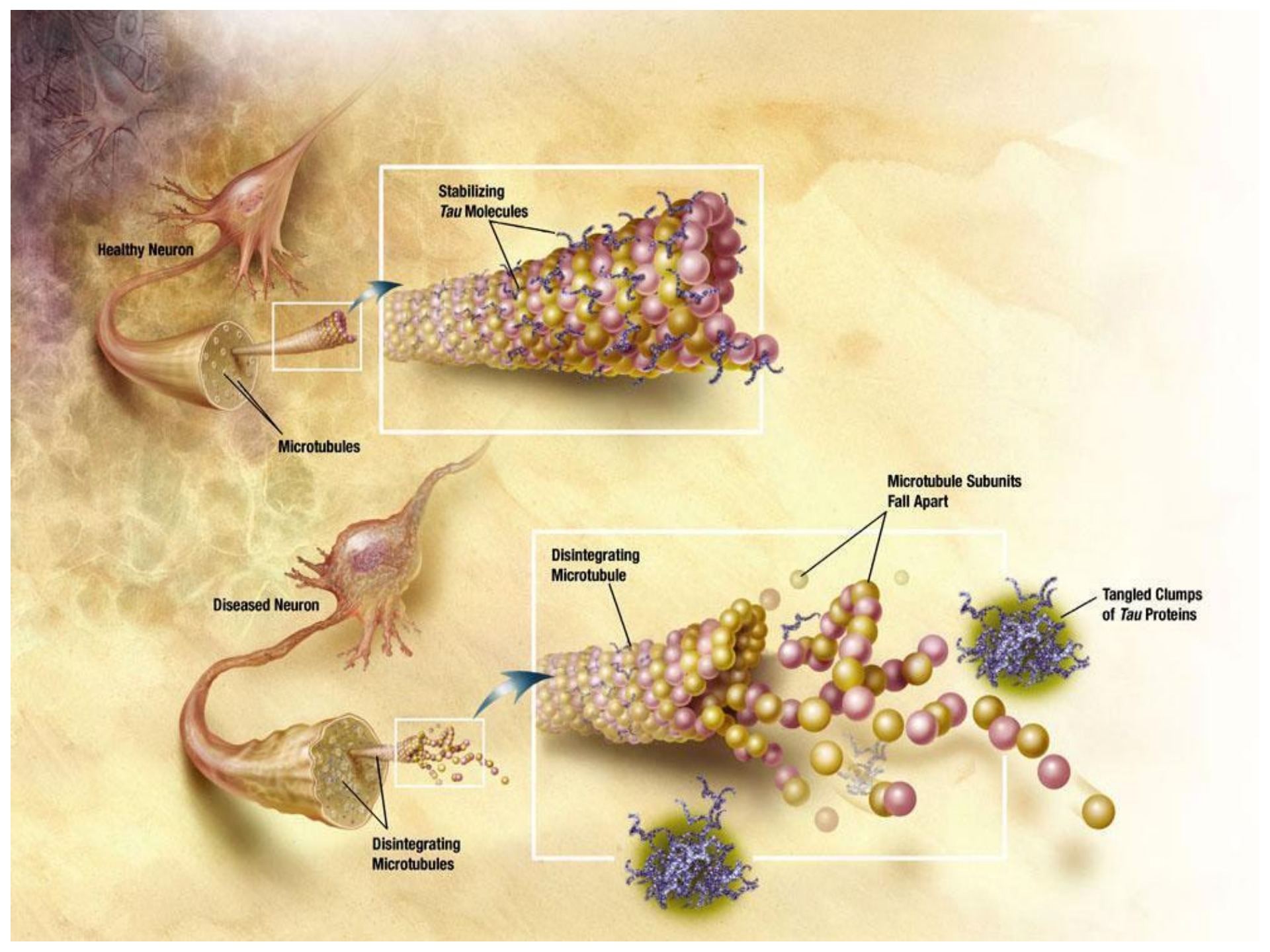
Diseased Neuron

Disintegrating
Microtubules

Disintegrating
Microtubule

Microtubule Subunits
Fall Apart

Tangled Clumps
of *Tau* Proteins



- The pathophysiologic process of AD begins well before clinical symptoms arise, and this period may be the optimal time to intervene if disease-modifying therapies can be identified.
- Laboratory and imaging biomarkers of AD are increasingly used in research settings.

Genetics involved in AD

- **Chromosome 21:**

β -amyloid precursor protein gene (APP) linked to **some familial AD (FAD)**

- **Chromosome 14:**

potential locus of **early-onset FAD**

- **Chromosome 19:**

potential locus of **late-onset AD**, both sporadic and familial

Carries the apolipoprotein E (ApoE) allele which may be one of several risk factor for AD. *

Diagnosis of AD

There is no exact diagnostic test or finding that makes AD unique.

The clinical diagnosis is “**Diagnosis by Exclusion,**” :

A) Definite AD

B) Probable AD

C) Possible AD

Definite Alzheimer's Disease

Clinical criteria present

+

Histopathologic evidence from brain biopsy(usually *post-mortem*) of:

- ✓ '**Senile plaques**' &
- ✓ '**Neurofibrillary tangles**' in *hippocampus* & *cerebral cortex*

Probable Alzheimer's Disease

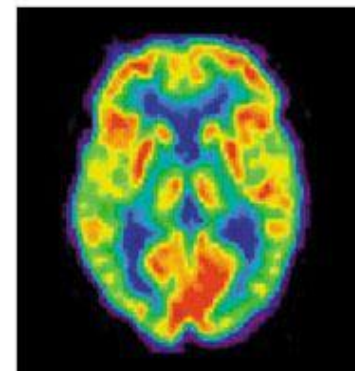
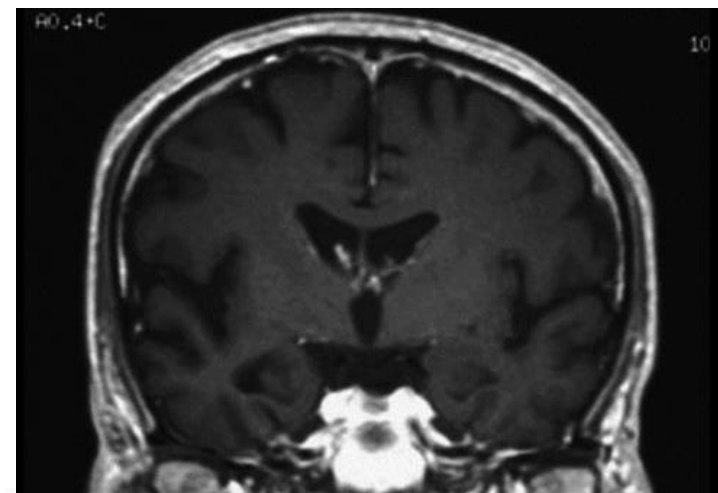
- Documentation of **cognitive impairment** with screening mental status examination, confirmed by neuropsychological testing
- Deficits in **two or more** areas of cognition
- Progressive worsening of memory & other cognitive function
- No disturbance of consciousness
- Onset between ages **40 - 90**, most often after 65
- Absence of other disorders.

Possible Alzheimer's Disease

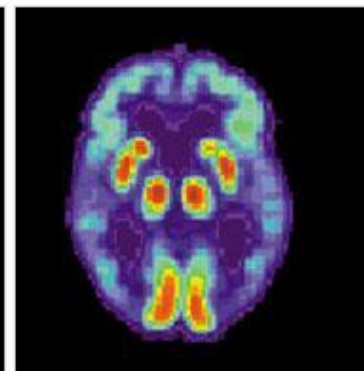
- There are **variations in the onset, presentation, or clinical course** of a dementing illness that are unusual in DAT but for which there is no alternative explanation
- There is a **second systemic or brain disease** capable of producing a dementia syndrome that is not considered to be the cause of the dementia in this case.
- There is a **single gradually progressive deficit**

Neuroimaging in AD

- Brain **MRI**s or **CT** scans can show diffuse cortical and/or cerebral atrophy, but these findings are not diagnostic of AD.
- **SPECT** or **PET** (see the image below) is not recommended for the routine workup of patients with typical presentations of AD.
- These modalities may be useful in atypical cases or when a form of fronto-temporal demntia is a more likely diagnosis.



PET Scan of Normal Brain



PET Scan of Alzheimer's Disease Brain

TREATMENT OF DEMEMENTIA

Reducing Risk of Dementia in Older Age

Ways to Reduce Your Risk of Dementia

Lifestyle habits

- Exercise regularly
- Eat a healthy diet
- Get adequate sleep
- Do not smoke

Health care

- Treat any heart disease
- Control your blood pressure
- Control your blood sugar
- Correct hearing loss

Mental and social well-being activities

- Stay curious and interested
- Actively learn new things
- Stay socially engaged
- Participate in group activities

A. Rekito

- **Treat reversible causes dementia:**

Like : depression, Hypothyroidism , B12 deficiency , --- etc. ---

- **Non-Pharmacological therapies:**

- Provide structured safe environment
- Close supervision when indicated
- Frequent orientation
- Familiar surroundings provide important cues - deteriorate rapidly when moved or hospitalized.
- If sudden worsening, evaluate for infection or side effect of new medication
- Caregiver support

Pharmacological therapies for AD:

No cure for AD

Therapies to improve dementia symptoms:

- * Enhancement of cholinergic system
- * Increase dietary choline: little effect on cognition
- * AChE inhibitors drugs: small cognitive enhancing effect

Therapies to slow progression:

- * **Memantine:** used in moderate to severe disease
- * **Vitamin E (antioxidant):**
2000 international units daily can be used in mild to moderate AD.
- * **Herbal remedies :**
- * **Other Experimental Therapies**

Symptomatic Treatment of dementia complications

Pharmacotherapy of AD:

- **Donepezil, rivastigmine, galantamine, and tacrine:**
are the drugs presently approved by the USA Food and Drug Administration (FDA) for treatment of:
 - AD. also for :
 - Dementia with Lewy bodies (DLB),
 - Vascular dementia (VaD), &
 - Parkinson disease (PD) dementia.
- Due to hepatotoxicity, **tacrine** is no longer used.
- The pharmacologic action of **donepezil, rivastigmine, and galantamine** is inhibition of cholinesterase, with a resulting increase in cerebral levels of acetylcholine.
- **Memantine** appears to act by blocking overexcited N-methyl-D-aspartate (**NMDA**) channels. FDA approved for treatment of moderate – severe AD.

Cholinesterase Inhibitors

- **Donepezil** (Aricept): used in mild to severe disease
- **Galantamine** (Razadyne): used in mild to moderate disease
- **Rivastigmine** (Exelon): used in mild to moderate disease
- Combination of more than one cholinesterase inhibitor is not recommended
- Switching can occur if patients are not tolerating the initial treatment or a treatment failure
- If MMSE decline is greater than 2-4 points in one year changing therapy is warranted

N-methyl-D Aspartate (NMDA) Antagonist (Memantine)

- Only one NMDA-antagonist is available
- Glutamate an excitatory neurotransmitter in the brain
- Blocks glutamatergic neurotransmission by antagonizing NMDA receptors
- Memantine is used in **moderate to severe disease**
- Not recommended in early stages of the disease



Symptomatic Treatment of AD

Secondary symptoms of AD (eg, depression, agitation, aggression, hallucinations, delusions, sleep disorders) can be problematic.

The following classes of psychotropic medications have been used :

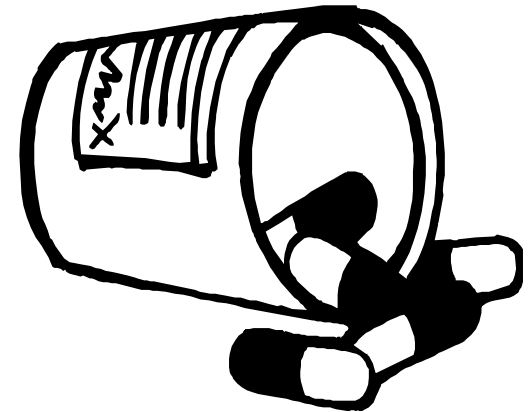
- Antidepressants
- Anxiolytics
- Antiparkinsonian agents
- Beta-blockers
- Antiepileptic drugs (for their effects on behavior)
- Neuroleptics

There are herbal remedies available.

- **Ginkgo biloba**
- **Phosphatidylserine**
- **Huperzine A:**

Highly purified derivative of Chinese cub moss

- **NB:** The use of ginkgo is not advocated because of questionable efficacy and lack of regulation, including variability in the dosing and contents of herbal extracts (inconsistent and unconvincing evidence of benefit).



Chinese cub moss

Experimental Therapies

In the past 10 years, numerous anti-amyloid therapy studies have been conducted to decrease toxic amyloid fragments in the brain, including studies of the following:

- Vaccination with amyloid species
- Administration of monoclonal anti-amyloid Abs.
- Administration of intravenous immune globulin that may contain amyloid-binding Abs.
- Selective amyloid-lowering agents
- Chelating agents to prevent amyloid polymerization
- Brain shunting to improve removal of amyloid
- Beta-secretase inhibitors to prevent generation of the A-beta amyloid fragment

Aducanumab:

- **Aducanumab** is a recombinant monoclonal antibody directed against amyloid beta.
- It is approved by the US - FDA for the treatment of mild AD.
- While aducanumab appears highly effective in reducing brain amyloid levels, it is uncertain that patients benefit clinically from treatment.

Aducanumab use

Indications :

- **Mild cognitive impairment or mild dementia** – MMSE) ≥ 21
- **Documented amyloid pathology** by PET or CSF studies.
- **No contraindications**

Dosing and administration

is administered by i.v. (IV) infusion every four weeks, typically in an infusion center.

Adverse Effects:

- **Amyloid Related Imaging Abnormalities** or "ARIA."

ARIA is a temporary swelling or small spots of bleeding in the brain that usually clears up with time. (seen on MRI)

- Hypersensitivity, Headache, falls, diarrhea, and confusion.

Take Home Message

- AD is an incurable disease with a **long preclinical period** and progressive course.
- Mental functions become **progressively more impaired** leading to profound dementia
- Brain MRI shows marked **brain atrophy** due to extensive neuronal loss.
- Diagnosis confirmed only by histology of post-mortem brain:
 - ‘**Senile plaques**’ &
 - ‘**Neurofibrillary tangles**’ in hippocampus & cerebral cortex
- There is no exact clinical test or finding that makes AD unique. “**Diagnosis by Exclusion,**”
- **No cure** for AD
- Limited therapy to help **symptoms** or to slow progression.
- **Memantine** is used in moderate to severe AD.
- **Cholinesterase Inhibitors:** are use in mild to severe AD, LBD, VaD & PD dementia.
- **Aducanumab** cab be used for mild cognitive impairment or mild dementia – MMSE) ≥ 21